

Digital Image Processing

Chapter 5: Morphological Image Processing

Preliminaries

- The language of mathematical morphology is set theory.
- In binary images, the set elements are members of the 2-D integer space Z^2, where each element (x,y) is a coordinate of a black (or white) pixel in the image.
- Grayscale digital images can be represented as sets whose components are in Z^3.
  - In this case, two components of each element of the set refer to the coordinates of a pixel, and the third corresponds to its discrete intensity value.

- Morphological operations are defined in terms of sets.
- In image processing, we use morphology with two types of sets of pixels:
  - **Objects** : sets of foreground pixels
  - **Structuring elements** (SE's) : can be specified in terms of both foreground and background pixels.
  - Structuring elements sometimes contain so-call "don't care" elements, denoted by ×, signifying that the value of that particular element in the SE does not matter. In this sense, the value can be ignored, or it can be made to fit a desired value in evaluation of an expression.

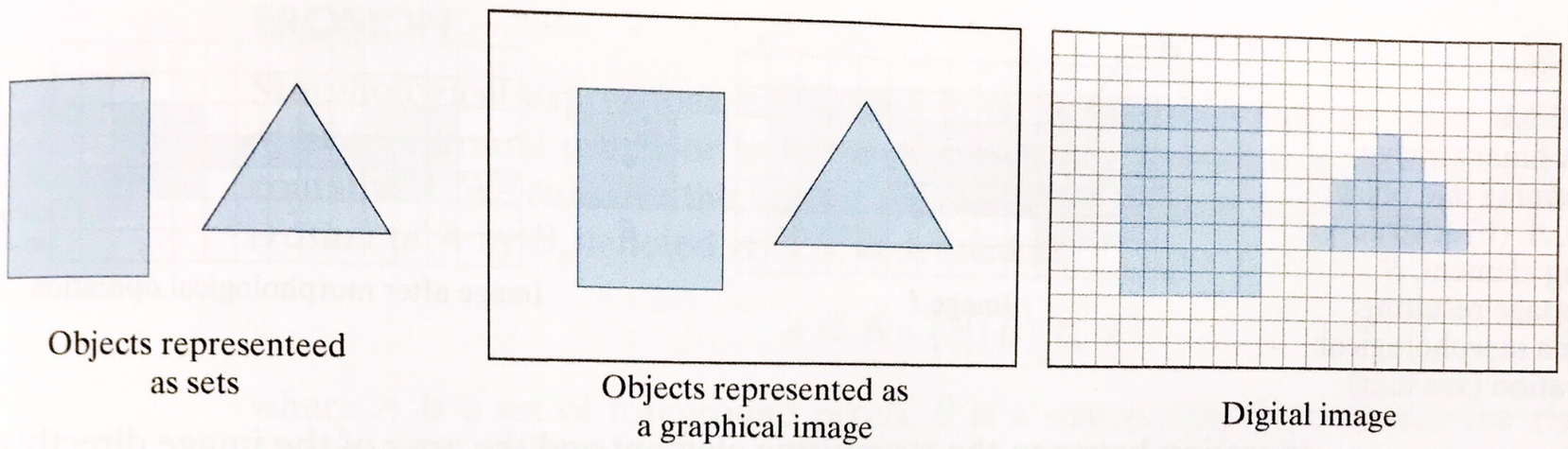


FIGURE 9.1 Top row. *Left:* Objects represented as graphical sets. *Center:* Objects embedded in a background to form a graphical image. *Right:* Object and background are digitized to form a digital image (note the grid). Second row: Example of a structuring element represented as a set, a graphical image, and finally as a digital SE.

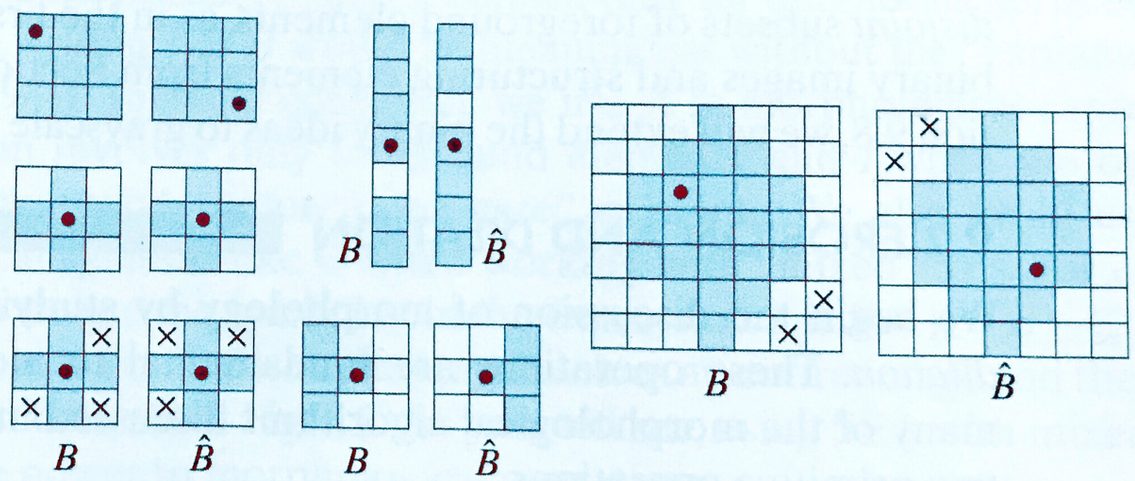
- There is an important difference between digital images and digital structuring elements.
- Observe on the top right that there is a border of background pixels surrounding the objects, while there is none in the SE.

Reflection

- The reflection of a set (structuring element) B about its origin, denoted by $\hat{B}$

$$\hat{B} = \{w | w = -b, \text{ for } b \in B\} \quad (1)$$

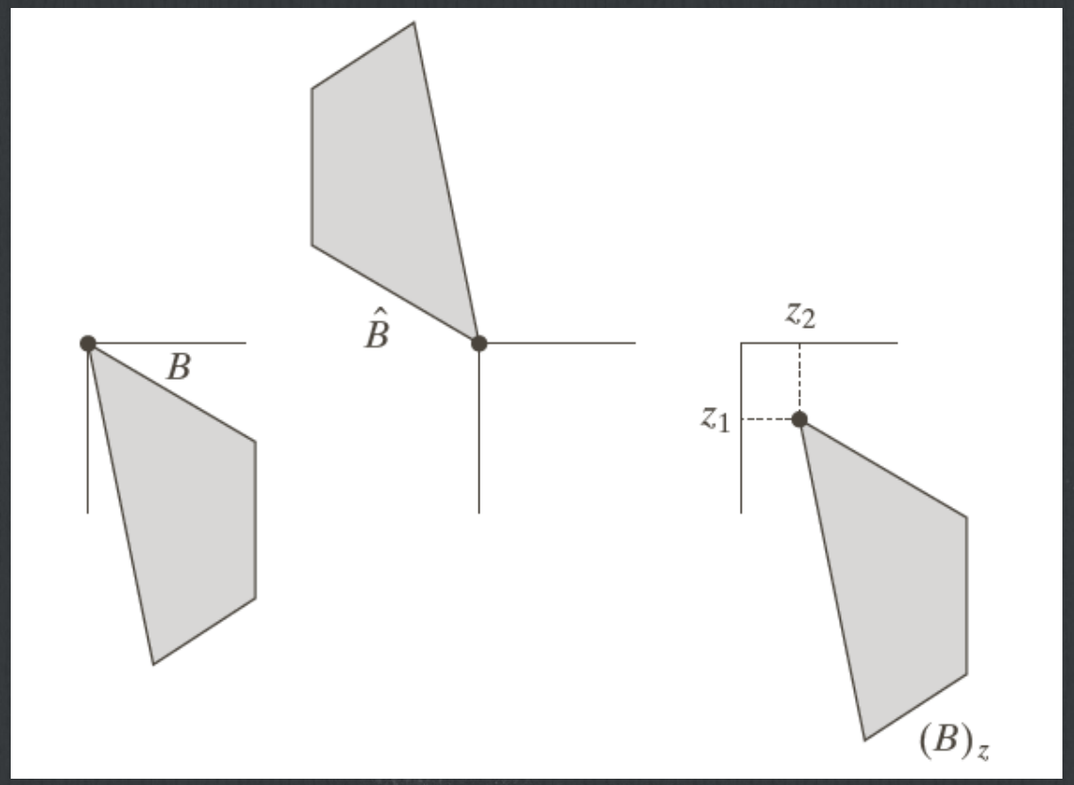
FIGURE 9.2
Structuring elements and their reflections about the origin (the ×'s are don't care elements, and the dots denote the origin). Reflection is rotation by 180° of an SE about its origin.



Translation

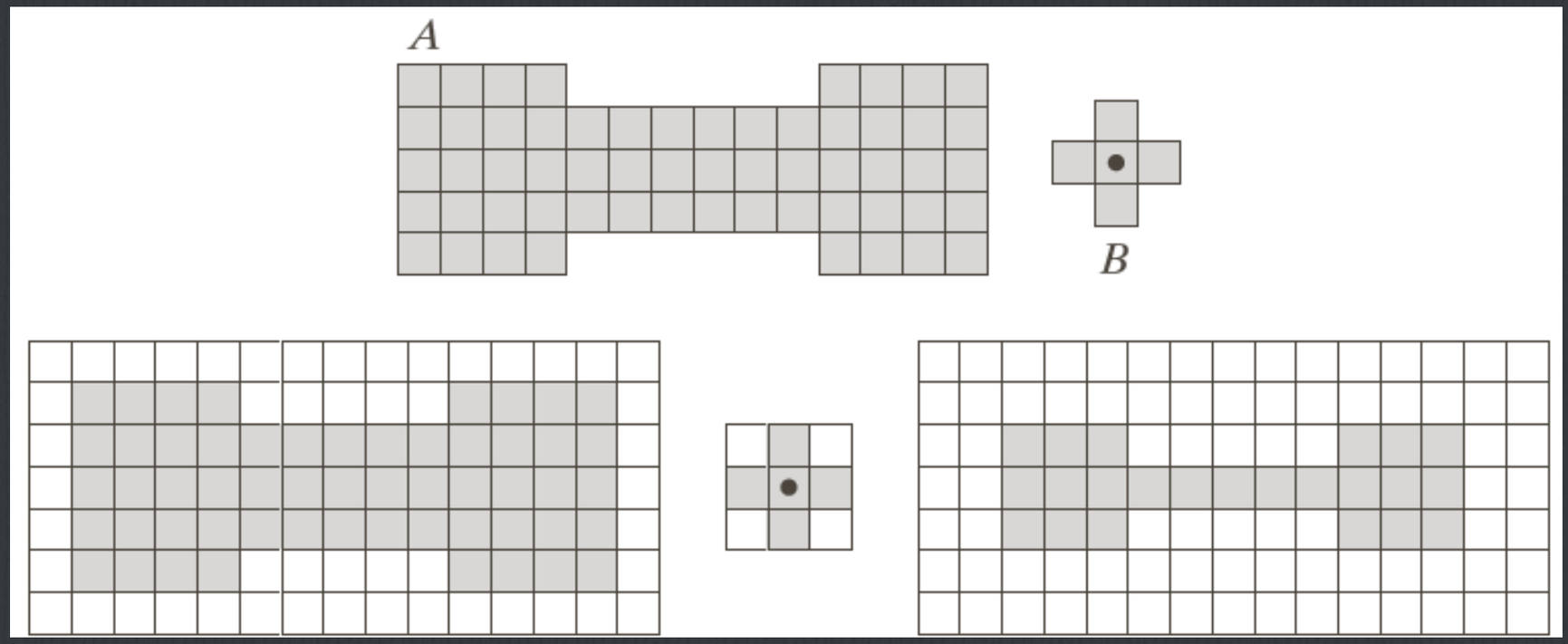
- The translation of a set B by point $z = (z_1, z_2)$ denoted $(B)_z$, is defined as

$$(B)_z = \{c | c = b + z, \text{ for } b \in B\} \tag{2}$$



a b c

FIGURE 9.1
(a) A set, (b) its reflection, and (c) its translation by z.



a b
c d e

FIGURE 9.3 (a) A set (each shaded square is a member of the set). (b) A structuring element. (c) The set padded with background elements to form a rectangular array and provide a background border. (d) Structuring element as a rectangular array. (e) Set processed by the structuring element.

Erosion and Dilation

Erosion

- A and B are sets in Z^2, the erosion of A by B, denoted $A \ominus B$

$$A \ominus B = \left\{ z \middle| (B)_z \subseteq A \right\} \quad (3)$$

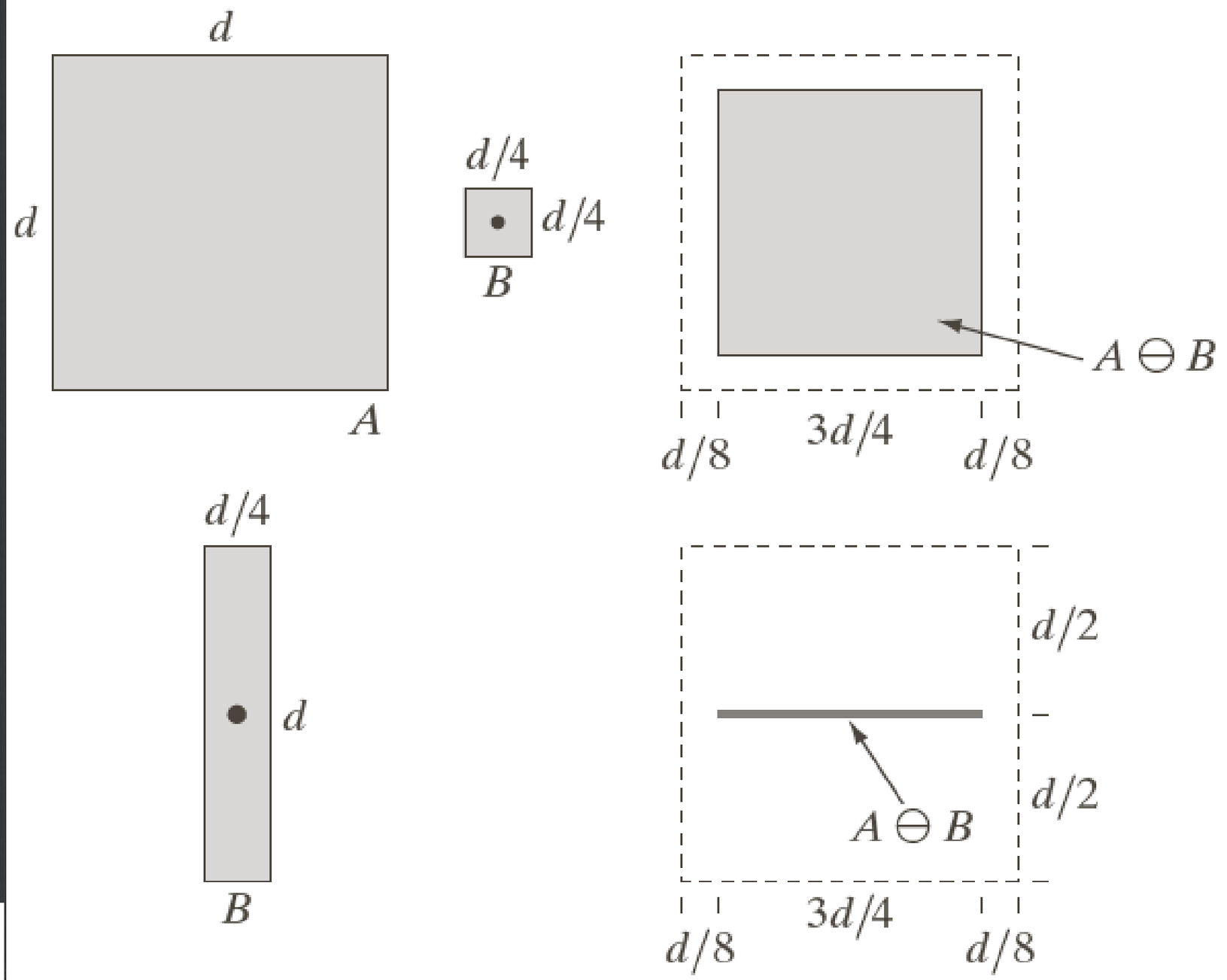
- A - set of foreground pixels
- B - structuring element
- z's - foreground values (1's)

- As noted, we work with see of foreground pixels embedded in a set of background pixels formed a complete image, I.

- Thus, inputs and outputs of our morphological procedures are images, not individual sets.

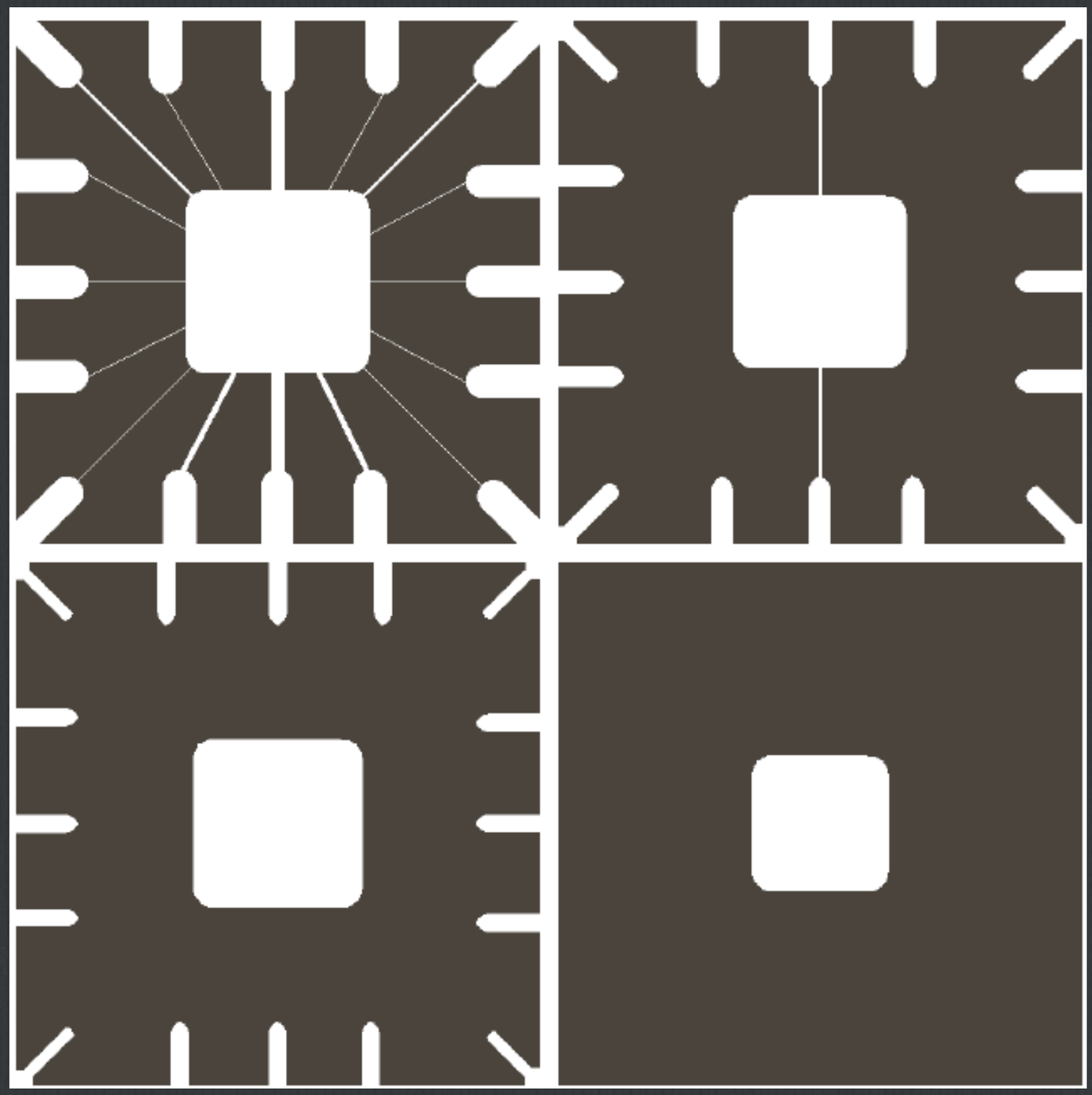
$$I \ominus B = \left\{ z \middle| (B)_z \subseteq A \text{ and } A \subseteq I \right\} \cup \left\{ A^c \middle| A^c \subseteq I \right\} \quad (4)$$

- I - rectangular array of foreground and background pixels.



a b c
d e

FIGURE 9.4 (a) Set A. (b) Square structuring element, B. (c) Erosion of A by B, shown shaded. (d) Elongated structuring element. (e) Erosion of A by B using this element. The dotted border in (c) and (e) is the boundary of set A, shown only for reference.



a b
c d

FIGURE 9.5 Using erosion to remove image components. (a) A 486 × 486 binary image of a wire-bond mask. (b)–(d) Image eroded using square structuring elements of sizes 11 × 11, 15 × 15, and 45 × 45, respectively. The elements of the SEs were all 1s.

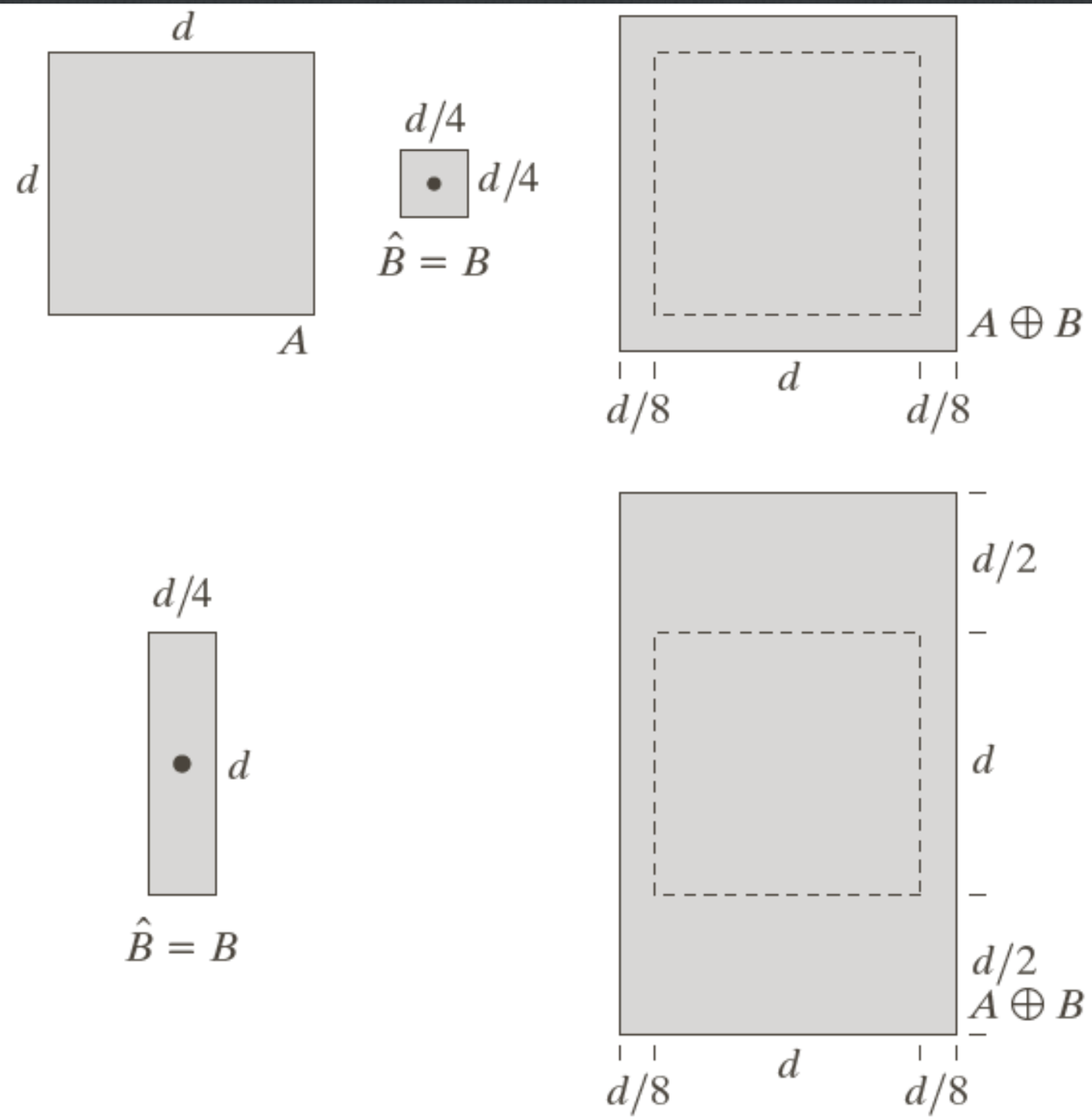
Dilation

- A and B are sets in Z^2, the dilation of A by B, denoted as $A \oplus B$

$$A \oplus B = \left\{ z \middle| \left(\hat{B} \right)_z \cap A \neq \varnothing \right\} \tag{4}$$

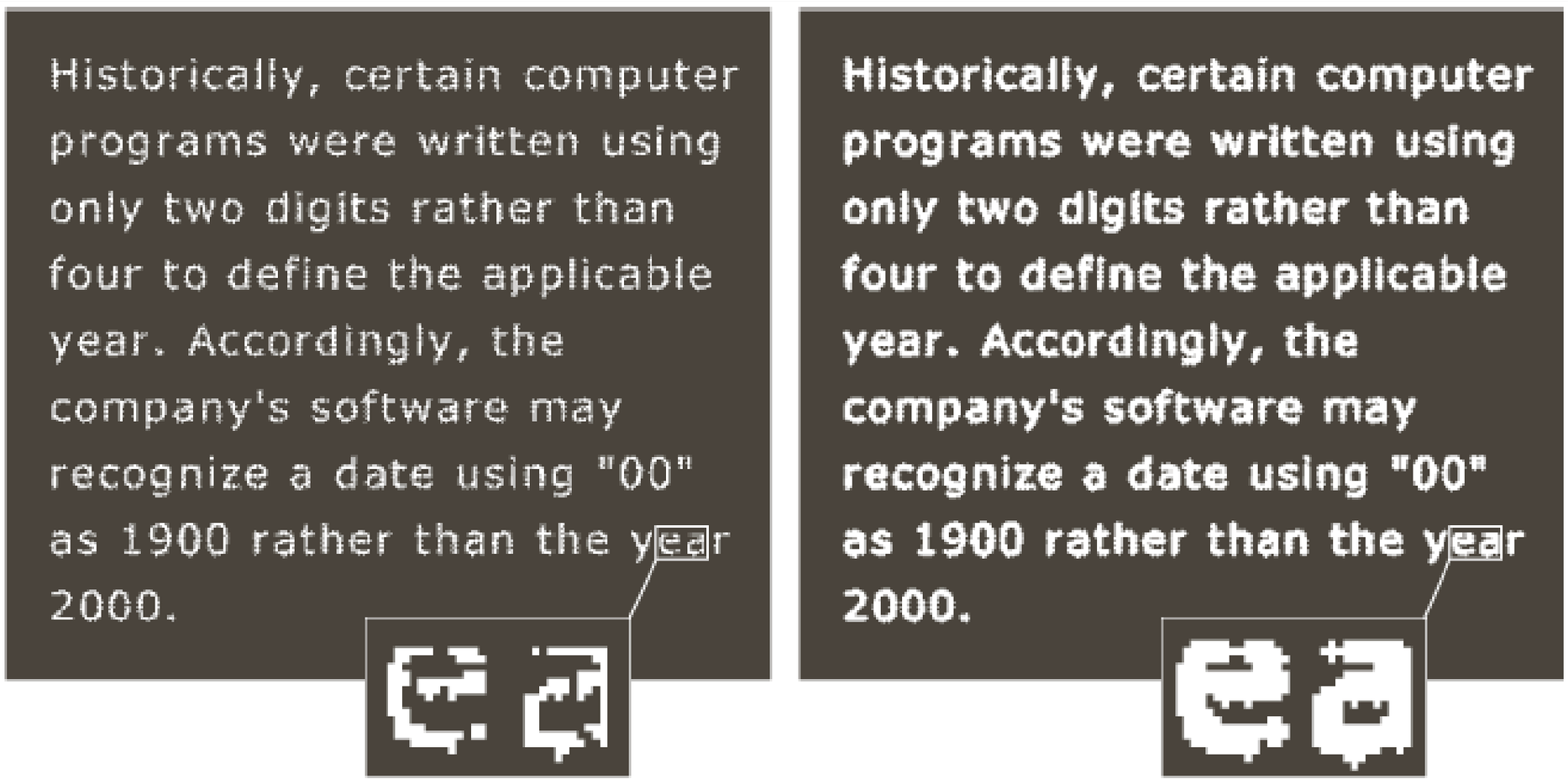
Or

$$A \oplus B = \left\{ z \middle| \left[\left(\hat{B} \right)_z \cap A \right] \subseteq A \right\} \tag{5}$$



a b c
d e

FIGURE 9.6
(a) Set A. (b) Square structuring element (the dot denotes the origin). (c) Dilation of A by B, shown shaded. (d) Elongated structuring element. (e) Dilation of A using this element. The dotted border in (c) and (e) is the boundary of set A, shown only for reference



a c
b

FIGURE 9.7
(a) Sample text of poor resolution with broken characters (see magnified view).
(b) Structuring element.
(c) Dilation of (a) by (b). Broken segments were joined.

0	1	0
1	1	1
0	1	0

Duality

- Erosion and dilation are **duals** of each other with respect to set complementation and reflection.

$$(A \ominus B)^c = A^c \oplus \hat{B} \tag{6}$$

And

$$(A \oplus B)^c = A^c \ominus \hat{B} \tag{7}$$

Opening and Closing

- In previous section, dilation expands the components of a set and erosion shrinks it.
- In this section,
  - Opening - generally smoothes the contour of an object, breaks narrow isthmuses, and eliminates thin protrusions.
  - Closing - tends to smooth sections of contours, but, as opposed to opening, it generally fuses narrow breaks and long thin gulfs, eliminates small holes, and fills gaps in the contour

Opening

- The **opening** of set A by structuring element B, denoted by A∘B

$$A \circ B = (A \ominus B) \oplus B \tag{8}$$

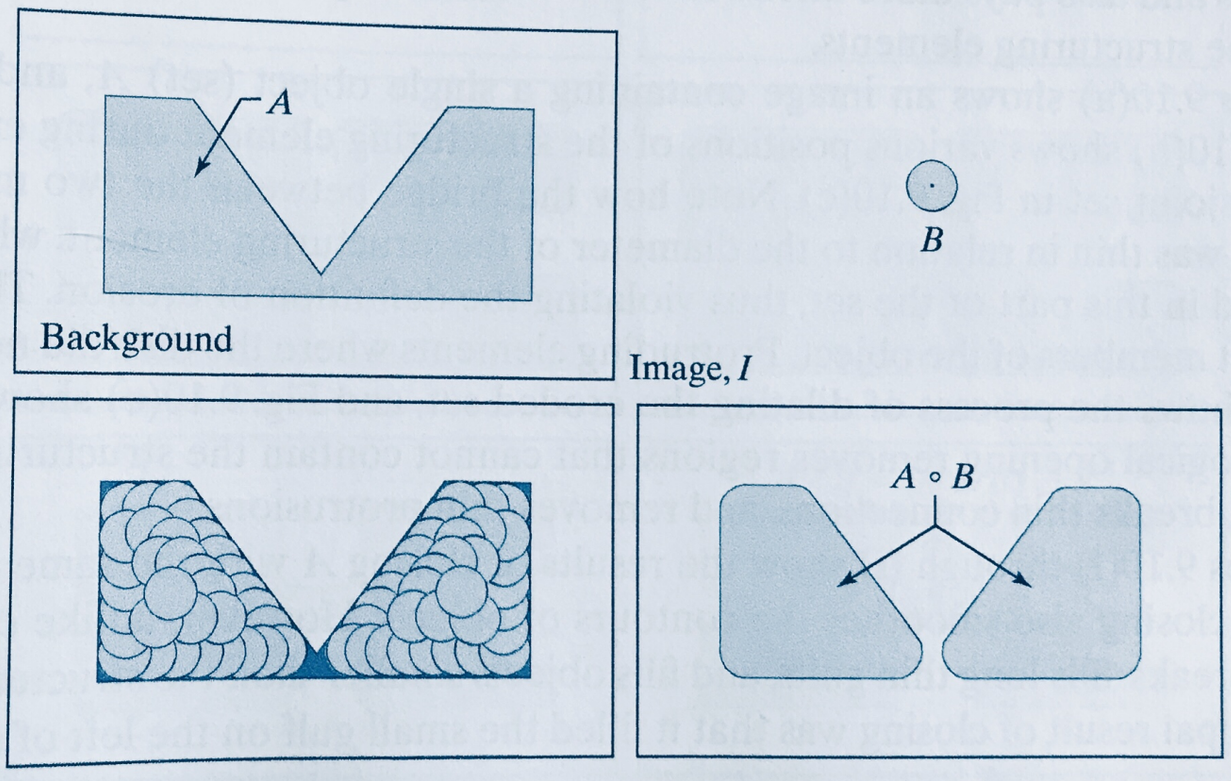
Closing

- The **closing** of A by B, denoted A•B

$$A \bullet B = (A \oplus B) \ominus B \tag{9}$$

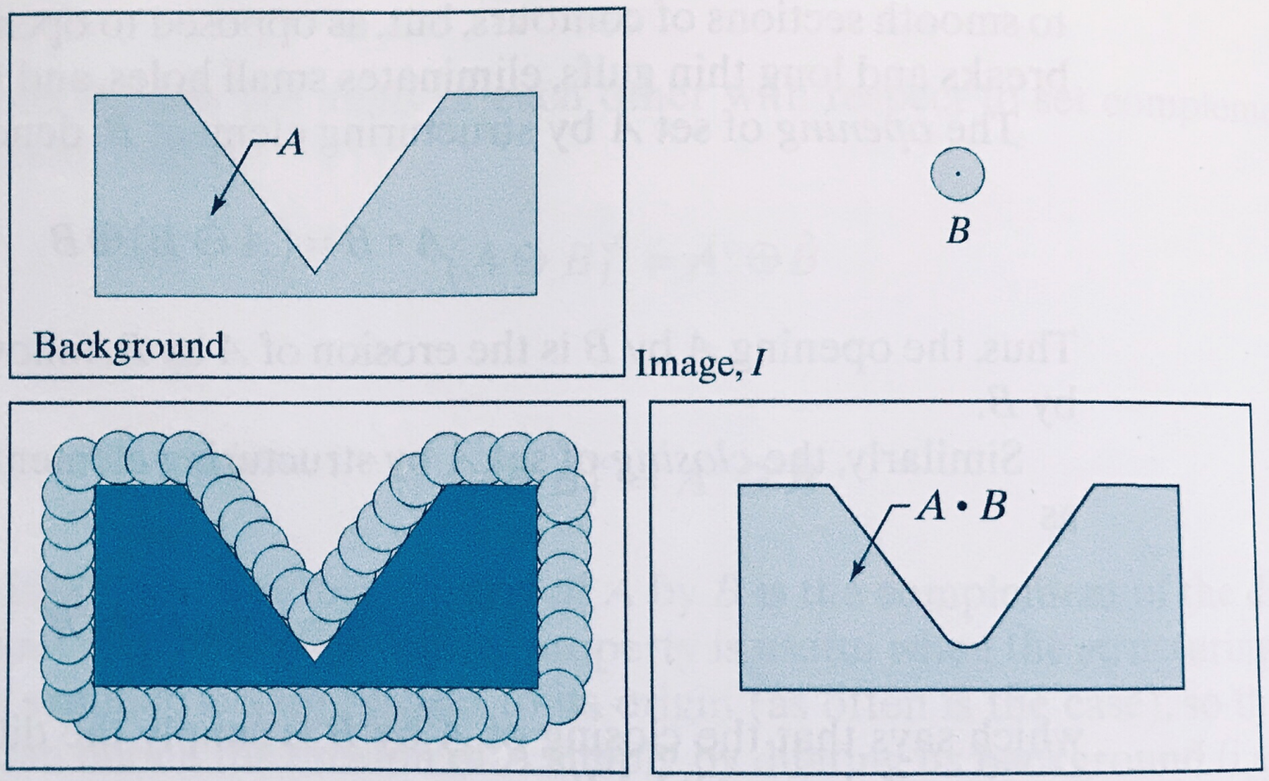
a b
c d

FIGURE 9.8
(a) Image I, composed of set (object) A and background.
(b) Structuring element, B.
(c) Translations of B while being contained in A. (A is shown dark for clarity.)
(d) Opening of A by B.



Opening

$$A \circ B = \bigcup \left\{ (B)_z \middle| (B)_z \subseteq A \right\} \tag{10}$$



a b
c d

FIGURE 9.9
(a) Image I, composed of set (object) A, and background. (b) Structuring element B. (c) Translations of B such that B does not overlap any part of A. (A is shown dark for clarity.) (d) Closing of A by B.

Closing

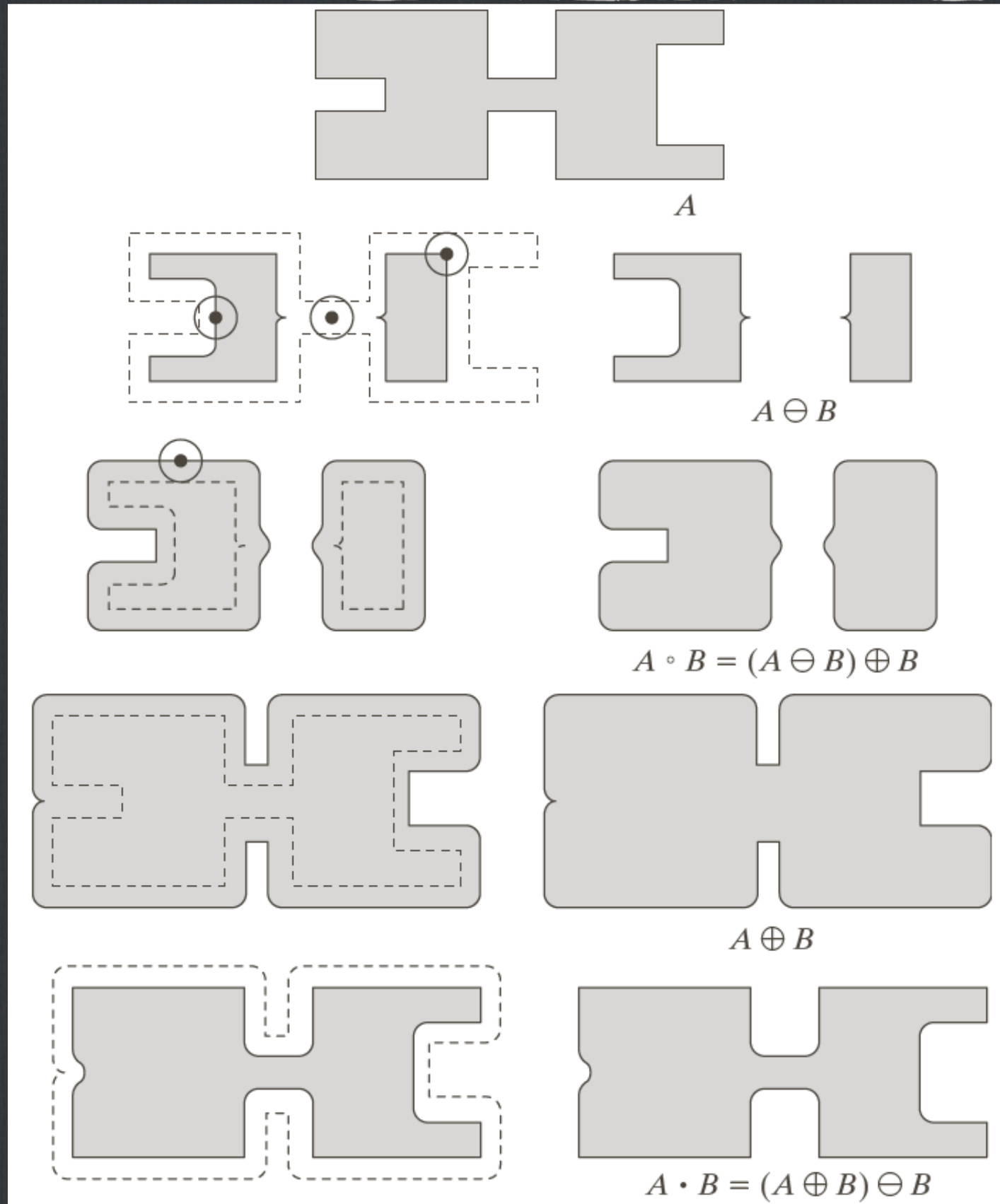
$$A \bullet B = \left[\cup \left\{ (B)_z \middle| (B)_z \cap A = \varnothing \right\} \right]^c \tag{11}$$

- As with erosion and dilation, opening and closing are **duals** of each other with respect to set complementation and reflection:

$$(A \circ B)^c = (A^c \bullet \hat{B}) \tag{12}$$

And

$$(A \bullet B)^c = (A^c \circ \hat{B}) \tag{13}$$



a
b c
d e
f g
h i

FIGURE 9.10 Morphological opening and closing. The structuring element is the small circle shown in various positions in (b). The SE was not shaded here for clarity. The dark dot is the center of the structuring element.

Morphological opening has the following properties:

(a) $A \circ B$ is a subset of A.

(b) If C is a subset of D, then $C \circ B$ is a subset of $D \circ B$.

(c) $(A \circ B) \circ B = A \circ B$

Morphological closing has the following properties:

(a) A is a subset of $A \bullet B$.

(b) If C is a subset of D, then $C \bullet B$ is a subset of $D \bullet B$.

(c) $(A \bullet B) \bullet B = A \bullet B$



a b
d c
e f

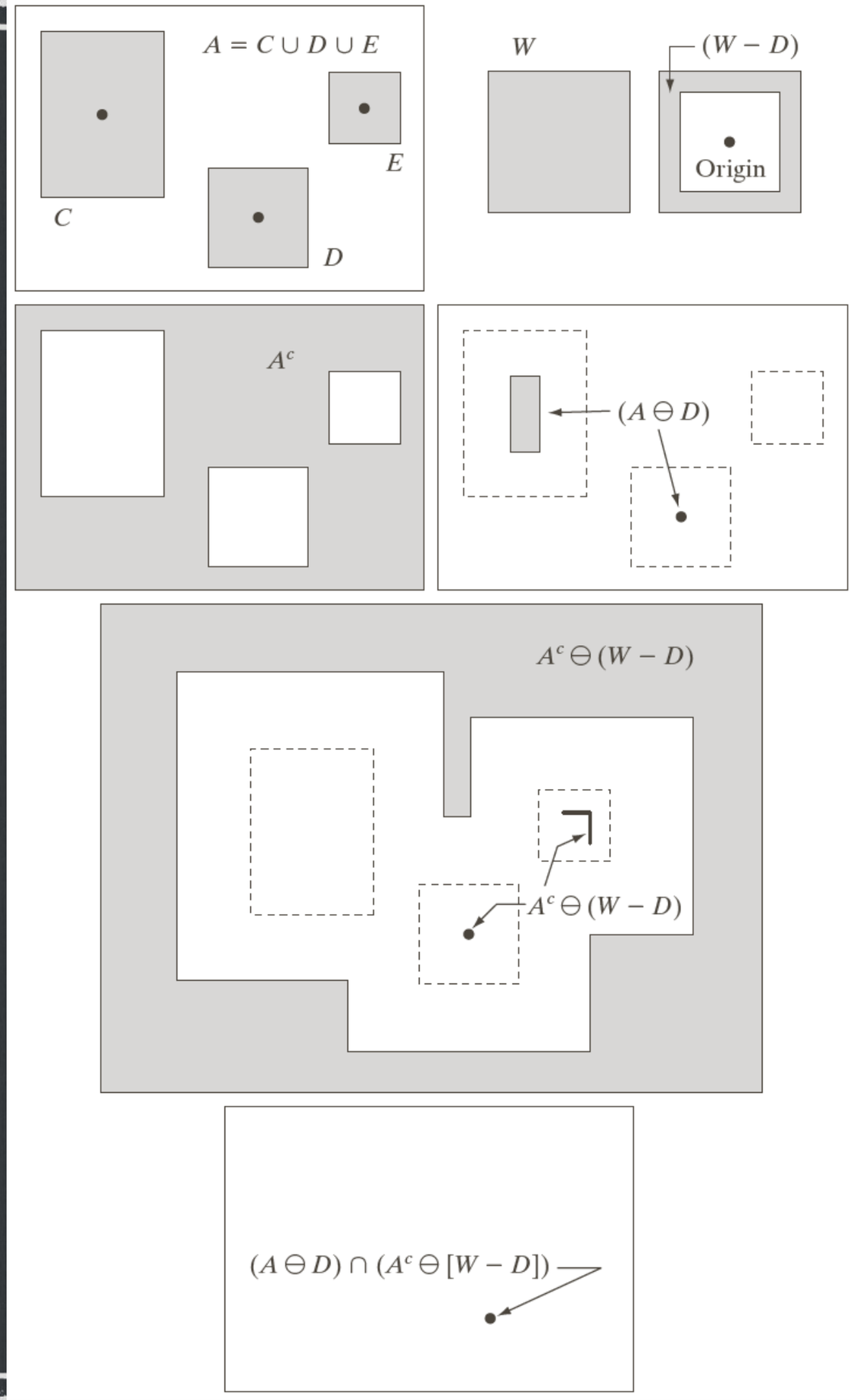
FIGURE 9.11
(a) Noisy image.
(b) Structuring element.
(c) Eroded image.
(d) Opening of A.
(e) Dilation of the opening.
(f) Closing of the opening.
(Original image courtesy of the National Institute of Standards and Technology.)

The Hit-or-Miss Transform

- The morphological **hit-or-miss transform (HMT)** is a basic tool for shape detection.
- I - a binary image composed of foreground (A) and background pixels, respectively.
- The HMT utilizes to structuring elements:
  - B_1 - for detecting shapes in the foreground
  - B_2 - for detecting shapes in the background
- The HMT of image I is defined as

$$I \circledast B_{1,2} = \left\{ z \middle| (B_1)_z \subseteq A \text{ and } (B_2)_z \subseteq A^c \right\}$$
$$= (A \ominus B_1) \cap (A^c \ominus B_2) \qquad (14)$$

- The morphological HMT is the set of translations, z, of structuring elements B_1 and B_2 such that, **simultaneously**,
  - B_1 found a match in the foreground (i.e., B_1 is contained in A)
  - B_2 found a match in in the background (i.e., B_2 is contained in A^c)
- The word "**simultaneous**" implies that z is the **same** translation of both structuring elements.
- The word "**miss**" in the HMT arises from the fact that B_2 finding a match in A^c is the same as B_2 not finding (missing) a match in A.



a b
c d
e
f

FIGURE 9.12
(a) Set A. (b) A window, W, and the local background of D with respect to W, $(W - D)$. (c) Complement of A. (d) Erosion of A by D. (e) Erosion of A^c by $(W - D)$. (f) Intersection of (d) and (e), showing the location of the origin of D, as desired. The dots indicate the origins of C, D, and E.

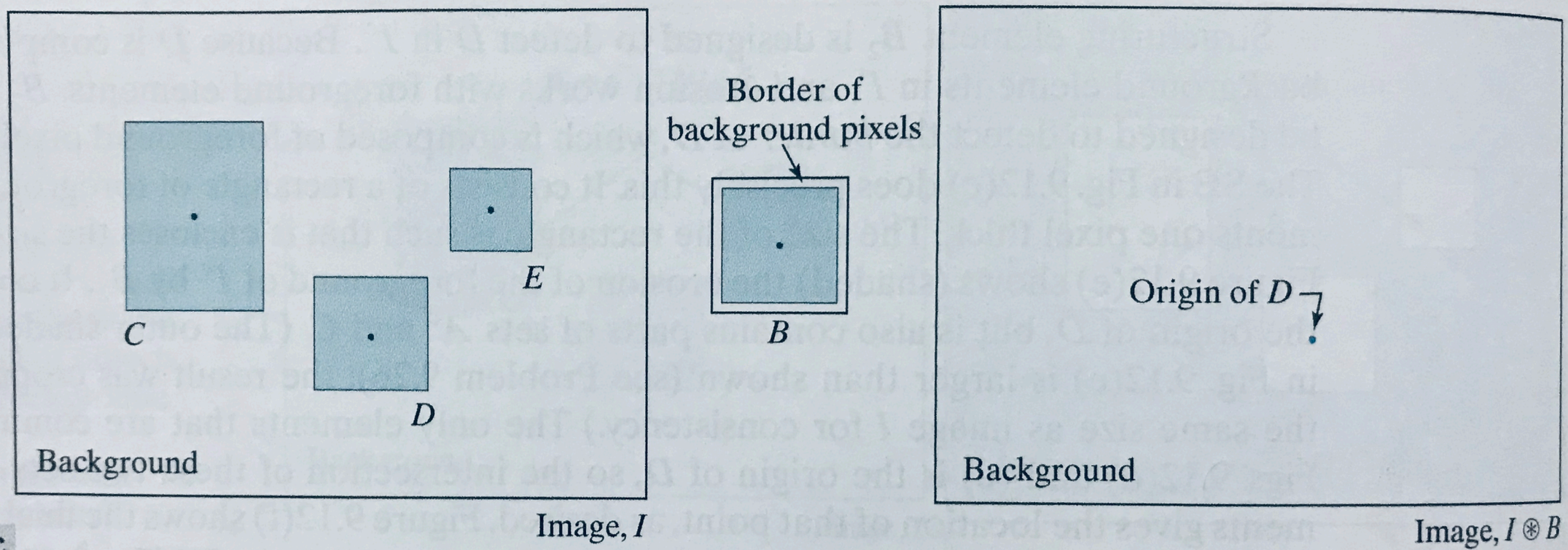
- Define a structuring element, B, identical to D, but having in addition a border of background elements with a width of one pixel.

- We can use a structuring element formed in such a way to restate the HMT as

$$I \circledast B = \left\{ z \middle| (B)_z \subseteq I \right\} \qquad (15)$$

- $(B)_z$ is a subset of image I, which is composed of both foreground and background pixels.

- B can be structured to detect any arrangement of pixels in image I.

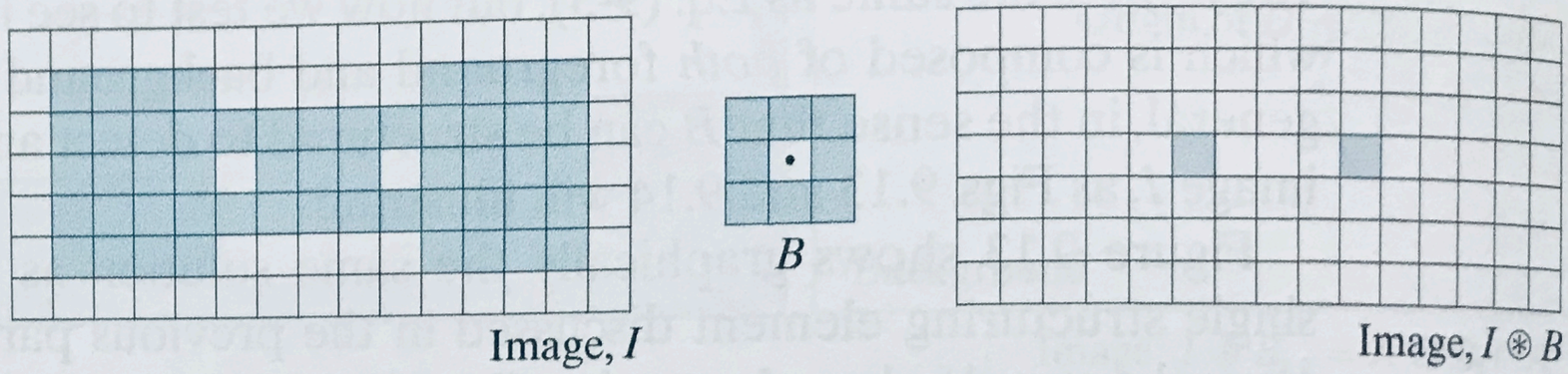


a b c

FIGURE 9.13 Same solution as in Fig. 9.12, but using Eq. (9-17) with a single structuring element.

a b c
d e f
g h i

FIGURE 9.14
Three examples of using a single structuring element and Eq. (9-17) to detect specific features. First row: detection of single-pixel holes. Second row: detection of an upper-right corner. Third row: detection of multiple features.



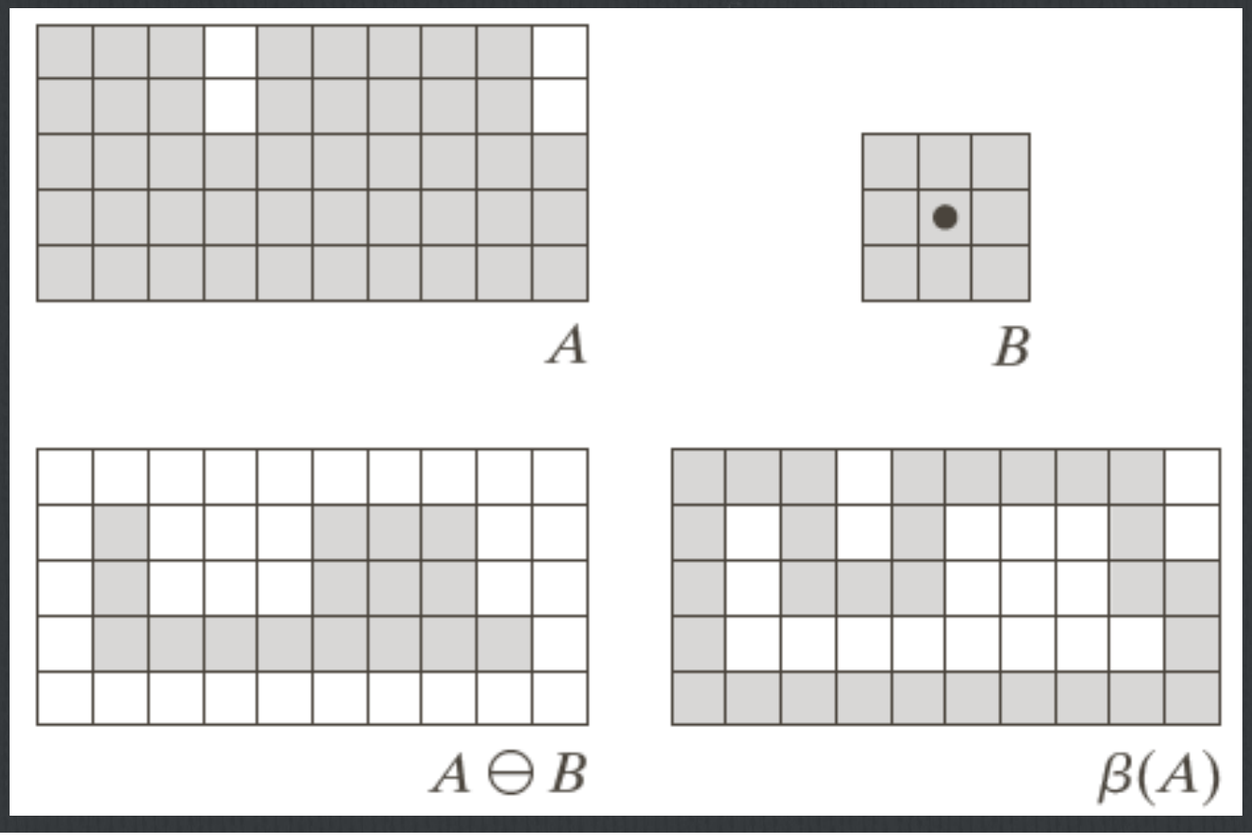
Some Basic Morphological Algorithms

Boundary Extraction

- β(A) - the boundary of a set A of foreground pixels

- β(A) can be obtained by first eroding A by a suitable structuring element B, and then performing the set difference between A and its erosion

$$\beta(A) = A - (A \ominus B) \tag{16}$$



a b
c d

FIGURE 9.13 (a) Set A. (b) Structuring element B. (c) A eroded by B. (d) Boundary, given by the set difference between A and its erosion.



a b

FIGURE 9.14
(a) A simple binary image, with 1s represented in white. (b) Result of using Eq. (9.5-1) with the structuring element in Fig. 9.13(b).

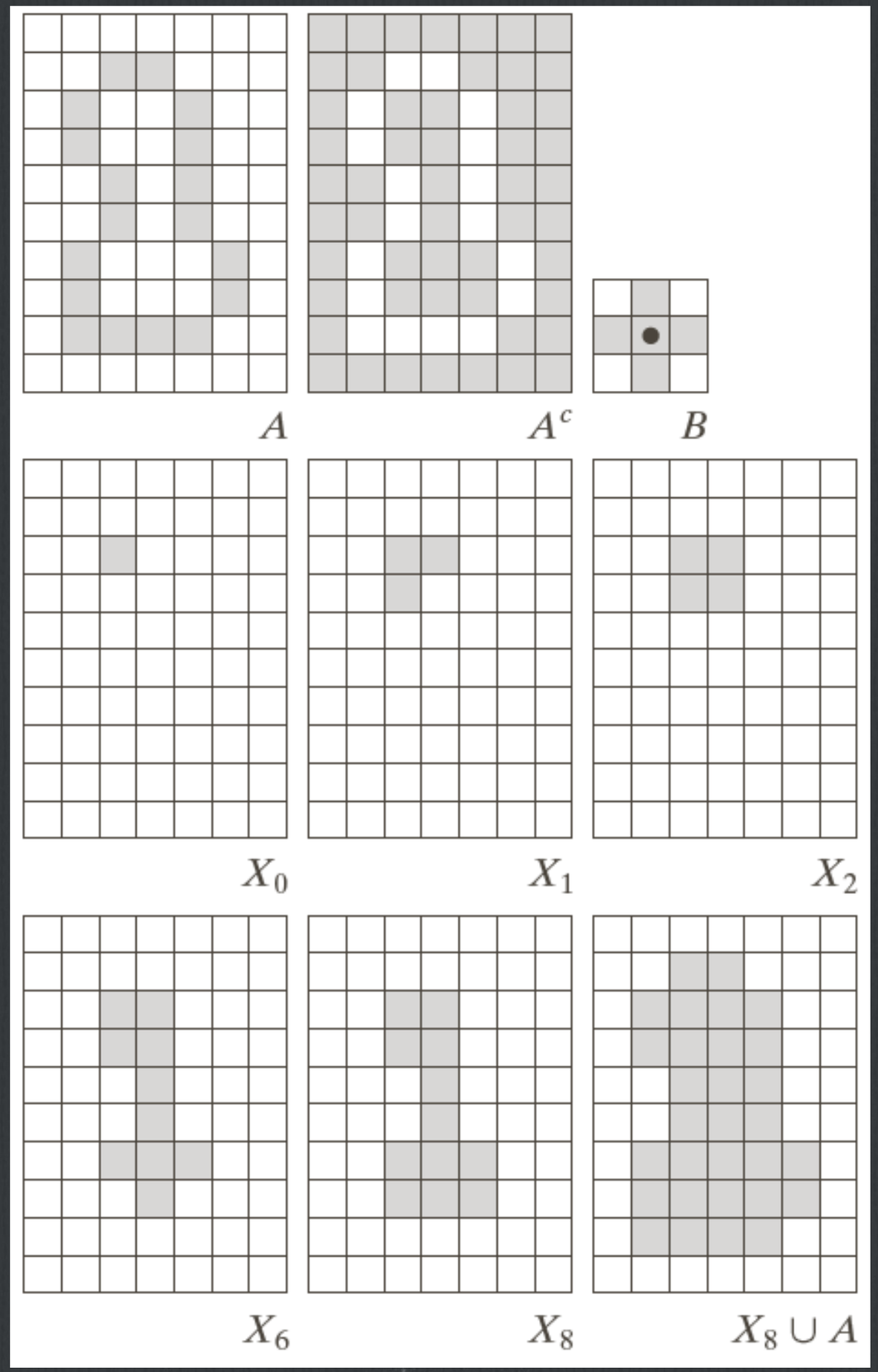
Hole Filling

- A hole may be defined as a background region surrounded by a connected border of foreground pixels.
- A - a set whose elements are 8-connected boundaries, with each boundary enclosing a background region (i.e., a hole)
- Given a point in each hole, the objective is to fill all holes with foreground elements (1’s)

- Begin by performing an array, X_0, of 0's (the same size as I, the image containing A)
- At locations in X_0 that correspond to pixels that are known to be holes, which we set to 1
- Then, the following procedure fills all holes with 1's:

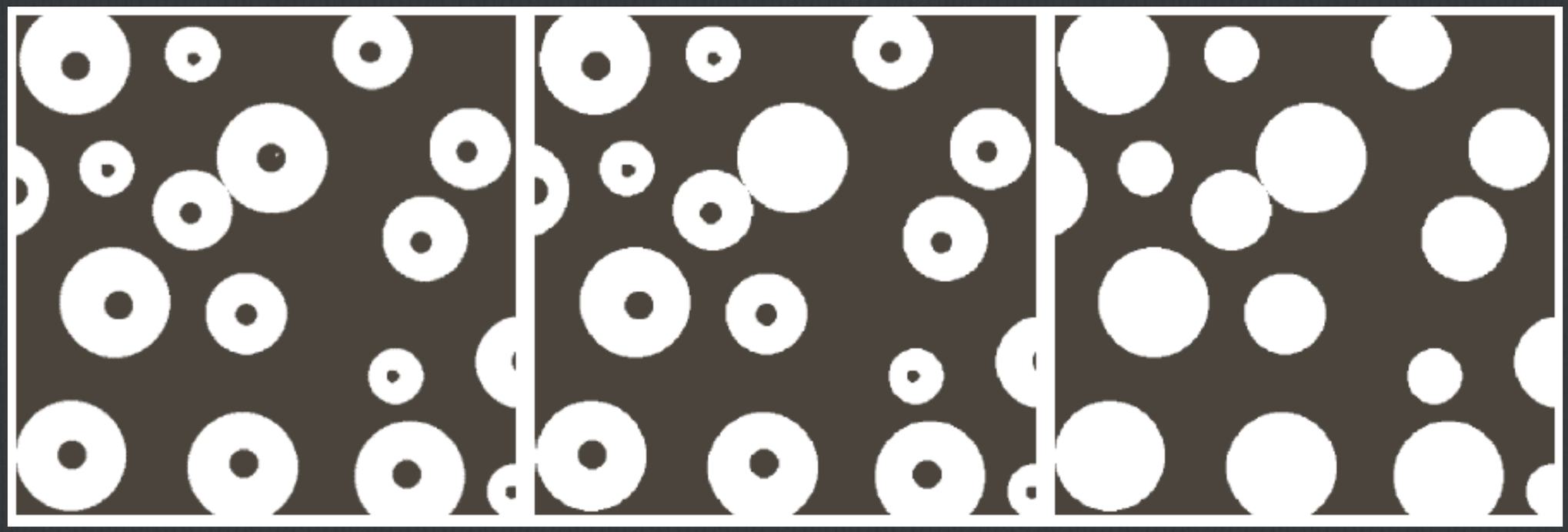
$$X_k = \left(X_{k-1} \oplus B\right) \cap I^c \qquad k = 1,2,3,... \tag{17}$$

- B - symmetric structuring element
- The algorithm terminates at iteration step k if $X_k = X_{k-1}$



a b c
d e f
g h i

FIGURE 9.15 Hole filling. (a) Set A (shown shaded). (b) Complement of A. (c) Structuring element B. (d) Initial point inside the boundary. (e)–(h) Various steps of Eq. (9.5-2). (i) Final result [union of (a) and (h)].



a b c

FIGURE 9.16 (a) Binary image (the white dot inside one of the regions is the starting point for the hole-filling algorithm). (b) Result of filling that region. (c) Result of filling all holes.

Extraction of Connected Components

- A - a set of foreground pixels consisting of one or more connected components

- Form an image X_0 (the same size as I, the image containing A) whose elements are 0's (background values),

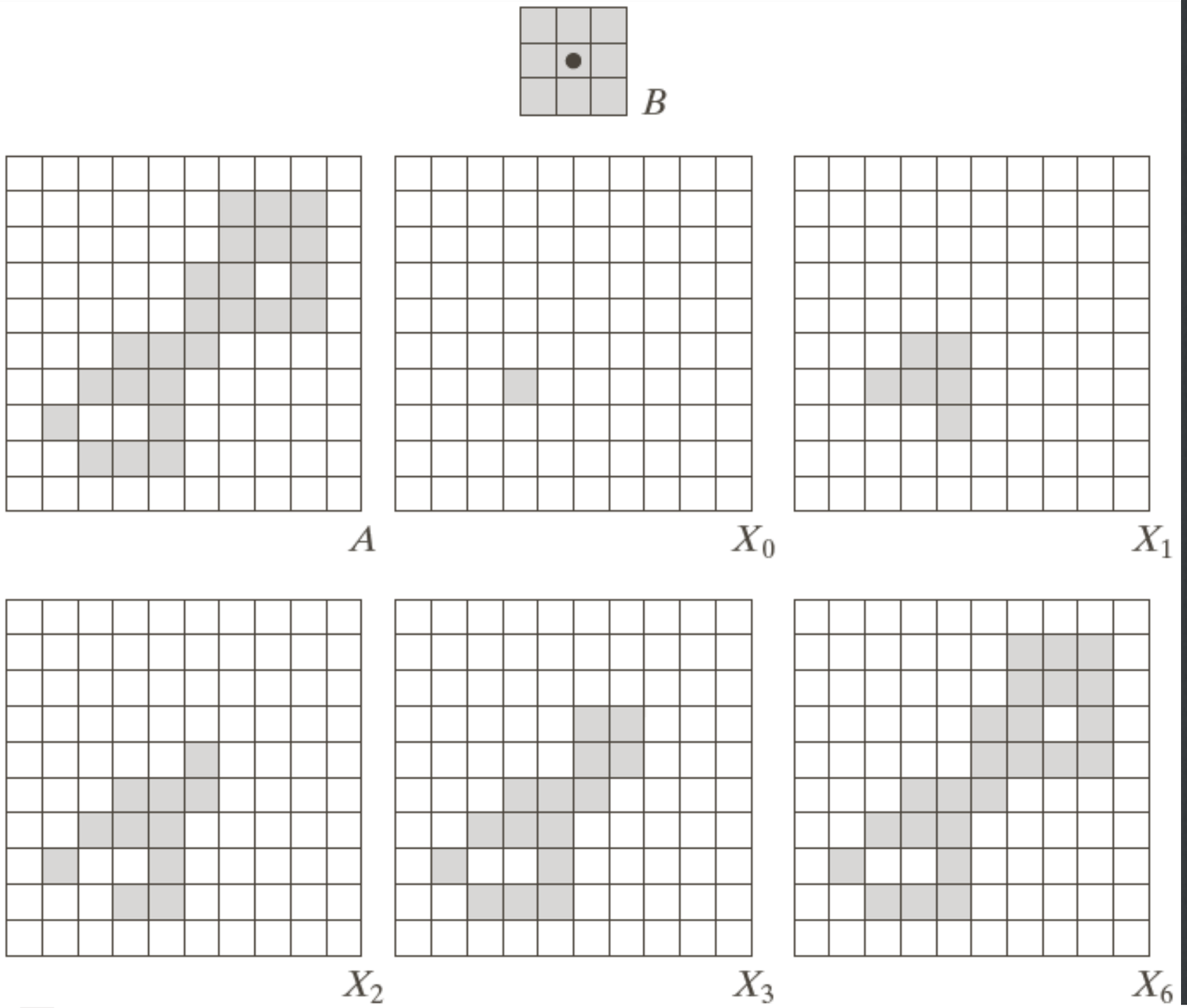
- except at each location known to correspond to a point in each component in A, which we set to 1 (foreground value)

- The objective is to start with X_0 and find all the connected components in I.

$$X_k = (X_{k-1} \oplus B) \cap I \qquad k = 1,2,3,... \tag{18}$$

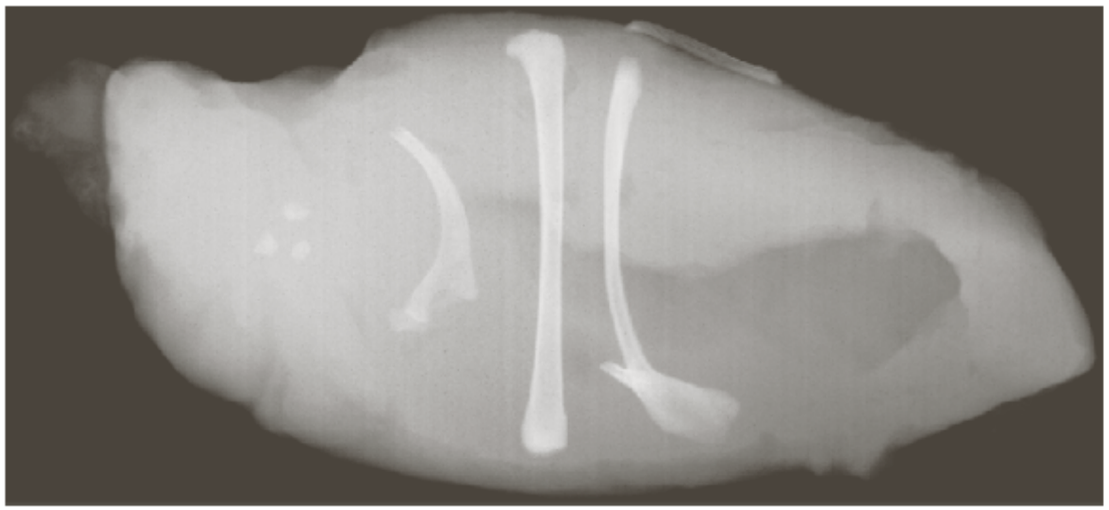
- B - structuring element

- The procedure terminates when $X_k = X_{k-1}$, with X_k containing all the connected components of foreground pixels in the image.



a
b c d
e f g

FIGURE 9.17 Extracting connected components. (a) Structuring element. (b) Array containing a set with one connected component. (c) Initial array containing a 1 in the region of the connected component. (d)–(g) Various steps in the iteration of Eq. (9.5-3).







Connected component	No. of pixels in connected comp
01	11
02	9
03	9
04	39
05	133
06	1
07	1
08	743
09	7
10	11
11	11
12	9
13	9
14	674
15	85

a
b
c d

FIGURE 9.18
(a) X-ray image of chicken filet with bone fragments. (b) Thresholded image. (c) Image eroded with a 5×5 structuring element of 1s. (d) Number of pixels in the connected components of (c). (Image courtesy of NTB Elektronische Geraete GmbH, Diepholz, Germany, www.ntbxray.com.)

Thinning

- The thinning of a set A by a structuring element B, denoted $A \otimes B$, can be defined in term of the hit-or-miss transform:

$$A \otimes B = A - (A \circledast B) \\ = A \cap (A \circledast B)^c \tag{19}$$

- A more useful expression for thinning A symmetrically is based on a sequence of structuring elements:

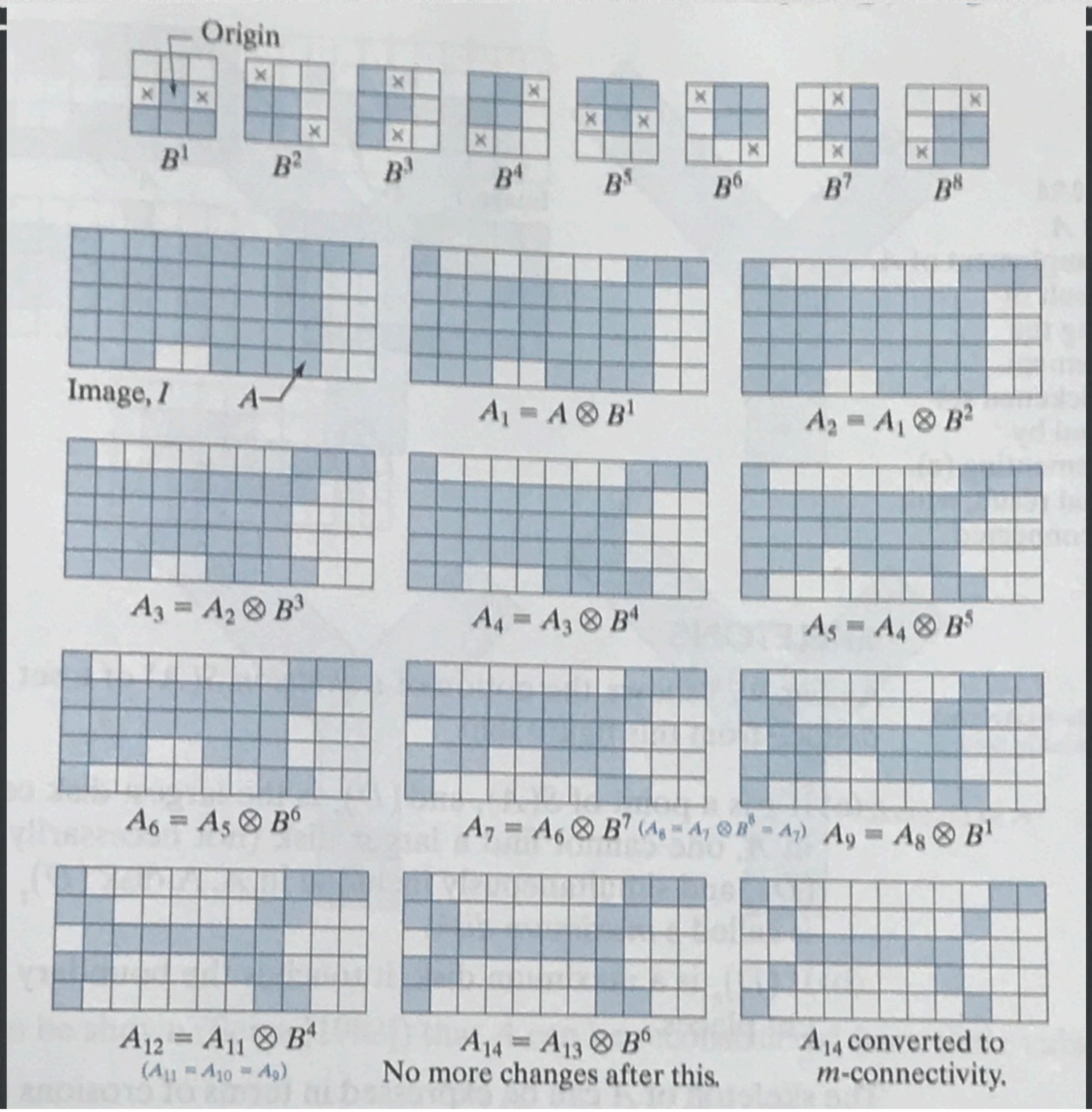
$$\{B\} = \{B^1, B^2, B^3, ..., B^n\} \tag{20}$$

- Thinning by a sequence of structuring elements:

$$A \otimes \{B\} = \left(\left(... \left(\left(A \otimes B^1 \right) \otimes B^2 \right) ... \right) \otimes B^n \right) \tag{21}$$

- The process is to thin A by one pass with B^1, then thin the result with one pass of B^2, and so on, until A is thinned with one pass of B^n.

- The entire process is repeated until no further changes occur after one complete pass through all structuring elements.



a
b c d
e f g
h i j
k l m

FIGURE 9.21 (a) Sequence of rotated structuring elements used for thinning. (b) Set A. (c) Result of thinning with the first element. (d)–(i) Results of thinning with the next seven elements (there was no change between the seventh and eighth elements). (j) Result of using the first four elements again. (l) Result after convergence. (m) Conversion to m-connectivity.

Thickening

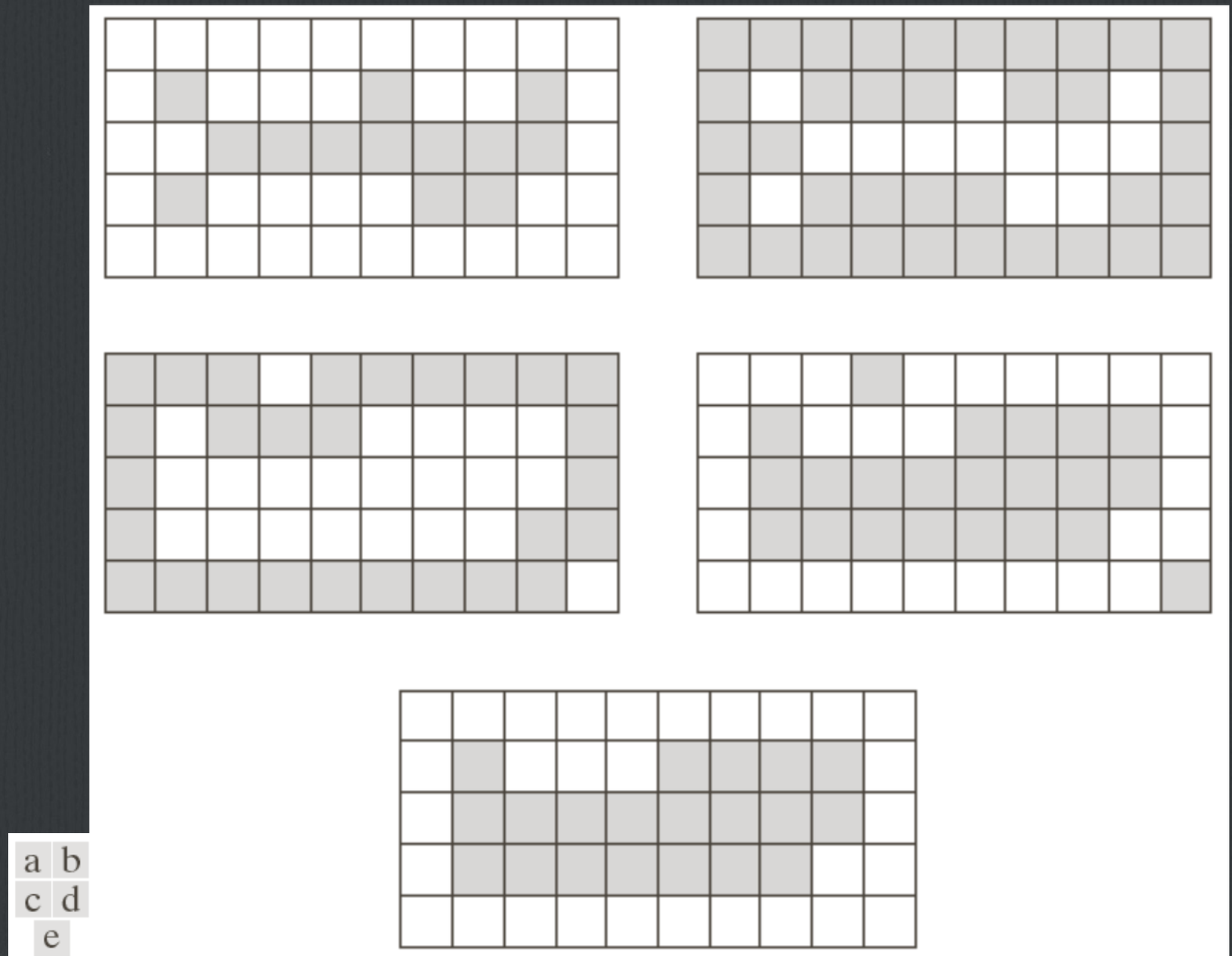
- Thickening is the morphological dual of thinning

$$A \odot B = A \cup (A \circledast B) \tag{22}$$

- B - structuring element suitable for thickening.

$$A \odot \{B\} = \left(\left(\ldots \left(\left(A \odot B^1 \right) \odot B^2 \right) \ldots \right) \odot B^n \right) \tag{23}$$

- The structuring elements used for thickening have the same form as those shown in Fig. 9.23(a), but with all 1's and 0' interchanged.



a b
c d
e

FIGURE 9.22 (a) Set A. (b) Complement of A. (c) Result of thinning the complement of A. (d) Thickened set obtained by complementing (c). (e) Final result, with no disconnected points.

Summary of Morphological Operations on Binary Images

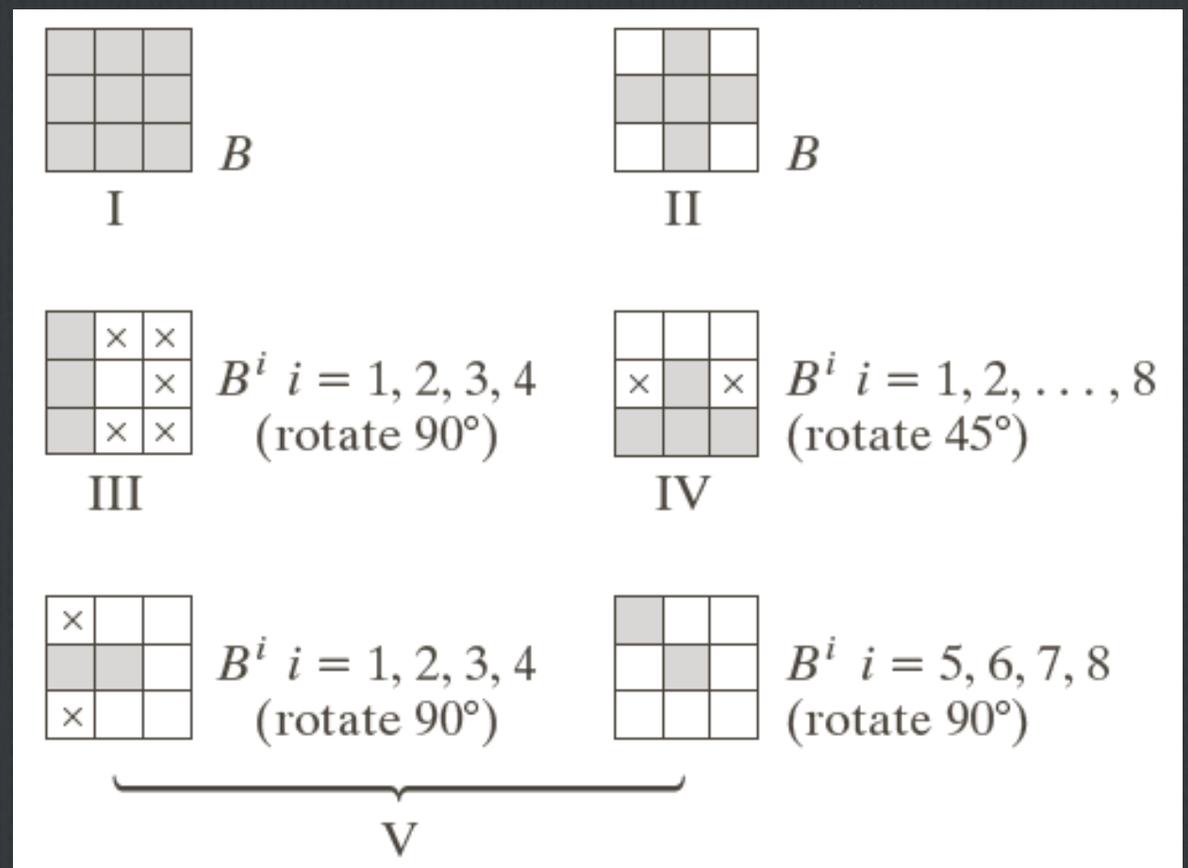


FIGURE 9.33 Five basic types of structuring elements used for binary morphology. The origin of each element is at its center and the ×'s indicate "don't care" values.

TABLE 9.1
Summary of morphological operations and their properties.

Operation	Equation	Comments (The Roman numerals refer to the structuring elements in Fig. 9.33.)
Translation	$(B)_z = \{w \mid w = b + z, \text{ for } b \in B\}$	Translates the origin of B to point z.
Reflection	$\hat{B} = \{w \mid w = -b, \quad \text{for } b \in B\}$	Reflects all elements of B about the origin of this set.
Complement	$A^c = \{w \mid w \notin A\}$	Set of points not in A.
Difference	$A - B = \{w \mid w \in A, w \notin B\} = A \cap B^c$	Set of points that belong to A but not to B.
Dilation	$A \oplus B = \left\{z \mid (\hat{B}_z) \cap A \neq \varnothing\right\}$	"Expands" the boundary of A. (I)
Erosion	$A \ominus B = \left\{z \mid (B)_z \subseteq A\right\}$	"Contracts" the boundary of A. (I)
Opening	$A \circ B = (A \ominus B) \oplus B$	Smoothes contours, breaks narrow isthmuses, and eliminates small islands and sharp peaks. (I)

(*Continued*)

TABLE 9.1 *(Continued)*

Operation	Equation	Comments (The Roman numerals refer to the structuring elements in Fig. 9.33.)
Closing	$A \bullet B = (A \oplus B) \ominus B$	Smoothes contours, fuses narrow breaks and long thin gulfs, and eliminates small holes. (I)
Hit-or-miss transform	$A \circledast B = (A \ominus B_1) \cap (A^c \ominus B_2)$ $= (A \ominus B_1) - (A \oplus \hat{B}_2)$	The set of points (coordinates) at which, simultaneously, B_1 found a match ("hit") in A and B_2 found a match in A^c
Boundary extraction	$\beta(A) = A - (A \ominus B)$	Set of points on the boundary of set A. (I)
Hole filling	$X_k = (X_{k-1} \oplus B) \cap A^c$; $k = 1, 2, 3, \ldots$	Fills holes in A; X_0 = array of 0s with a 1 in each hole. (II)
Connected components	$X_k = (X_{k-1} \oplus B) \cap A$; $k = 1, 2, 3, \ldots$	Finds connected components in A; X_0 = array of 0s with a 1 in each connected component. (I)
Convex hull	$X_k^i = (X_{k-1}^i \circledast B^i) \cup A$; $i = 1, 2, 3, 4$; $k = 1, 2, 3, \ldots$; $X_0^i = A$; and $D^i = X_{\text{conv}}^i$	Finds the convex hull $C(A)$ of set A, where "conv" indicates convergence in the sense that $X_k^i = X_{k-1}^i$. (III)
Thinning	$A \otimes B = A - (A \circledast B)$ $= A \cap (A \circledast B)^c$ $A \otimes \{B\} =$ $((\ldots((A \otimes B^1) \otimes B^2)\ldots) \otimes B^n)$ $\{B\} = \{B^1, B^2, B^3, \ldots, B^n\}$	Thins set A. The first two equations give the basic definition of thinning. The last equations denote thinning by a sequence of structuring elements. This method is normally used in practice. (IV)
Thickening	$A \odot B = A \cup (A \circledast B)$ $A \odot \{B\} =$ $((\ldots(A \odot B^1) \odot B^2 \ldots) \odot B^n)$	Thickens set A. (See preceding comments on sequences of structuring elements.) Uses IV with 0s and 1s reversed.
Skeletons	$S(A) = \bigcup_{k=0}^{K} S_k(A)$ $S_k(A) = \bigcup_{k=0}^{K} \{(A \ominus kB)$ $- [(A \ominus kB) \circ B]\}$ Reconstruction of A: $A = \bigcup_{k=0}^{K} (S_k(A) \oplus kB)$	Finds the skeleton $S(A)$ of set A. The last equation indicates that A can be reconstructed from its skeleton subsets $S_k(A)$. In all three equations, K is the value of the iterative step after which the set A erodes to the empty set. The notation $(A \ominus kB)$ denotes the kth iteration of successive erosions of A by B. (I)

(*Continued*)